Vendor Performance Analysis

Inventory, margin and vendor analysis for a liquor and wine distributor · 126 vendors · 10,692 brands · Afeef Shaikh

> **Headline:** $6.9M of inventory, at cost, sits in 2,613 slow-moving brands that sold less than 80% of what was bought during the period. Martignetti, Ultra Beverage and Perfecta Wines hold $2.9M of it, so reorder quantities with these three vendors are the first place to recover cash.

Business Problem

- Which vendors drive sales and gross profit, and how concentrated is purchasing?
- Where is capital tied up in stock that isn't selling?
- Does buying in bulk lower unit costs?
- How does freight cost vary by vendor?
- Do margins differ between top-selling and low-selling brands?

Key Metrics

Metric	Value	Definition
Total revenue	$451.62M	All 10,514 brands with sales
Gross profit	$138.47M	Revenue minus cost of the units actually sold
Gross margin	30.7%	Revenue-weighted
Freight cost	$1.64M	0.51% of purchase cost; 0.36% of revenue
Slow-moving stock	$6.94M	Unsold units at cost in brands with turnover below 0.8

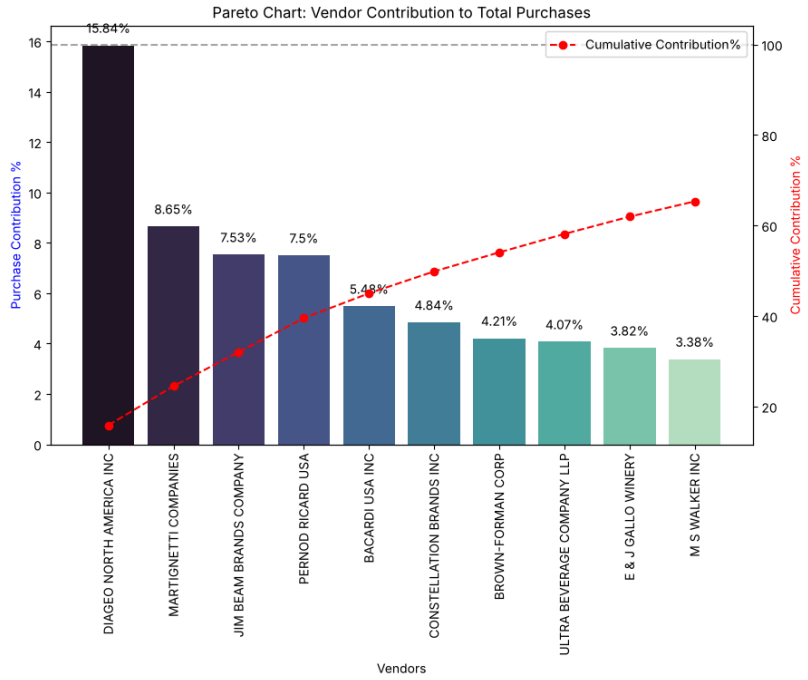
Data Quality Corrections

An earlier version of this analysis reported freight of $543M, more than total revenue. Tracing that figure uncovered three pipeline errors, all corrected in this report.

Issue	Cause	Correction
Freight counted repeatedly	Vendor-level freight was copied onto every brand of that vendor, then summed across brands.	Freight allocated to brands by share of purchase dollars: $543.4M → $1.64M.
Unsold stock treated as a loss	Gross profit was sales minus total purchases; 1,949 brands were wrongly flagged as unprofitable and excluded.	Gross profit = sales − units sold × unit cost. 104 brands are truly sold below cost.
Capital-at-risk formula	Total unsold units × the average purchase price of all brands.	Each brand's unsold units × its own price: $18.63M → $6.94M.

1. Vendor Concentration

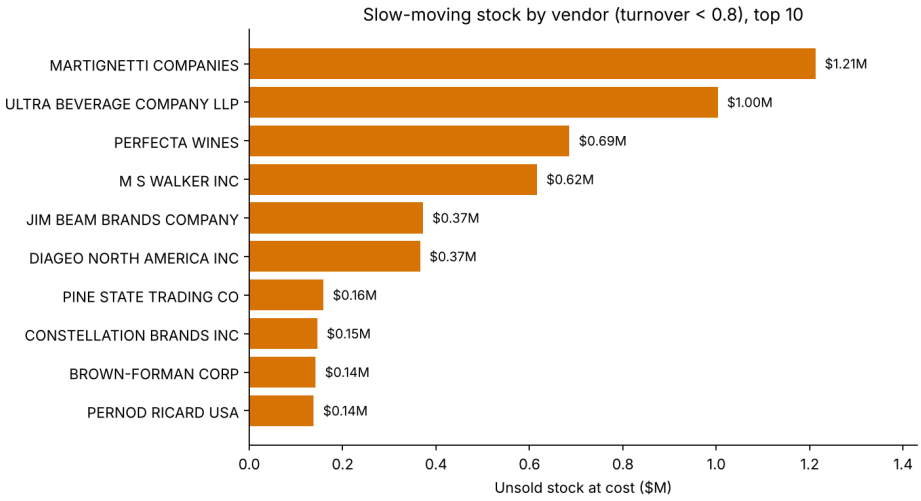
The top 10 vendors supply **65.3%** of purchase dollars, and Diageo alone supplies 15.8%. A supply disruption at one of the top four would affect between 7.5% and 15.8% of purchasing.



Recommendation: keep secondary suppliers qualified for the top-selling brands of the four largest vendors.

2. Capital Tied Up in Slow-Moving Stock

Purchases outran sales by $15.6M at cost in the brands that bought more than they sold ($8.7M net once brands that drew stock down are included). The part worth acting on is the **$6.94M** in 2,613 slow-moving brands, 178 of which sold nothing at all ($0.28M).



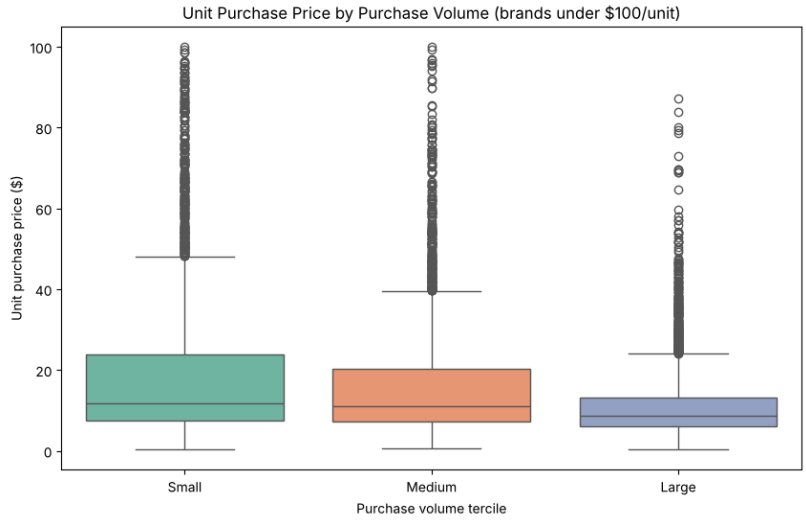
Recommendation: review reorder points with Martignetti ($1.21M), Ultra Beverage ($1.00M) and Perfecta Wines ($0.69M) first. *Caveat:* unsold stock is estimated as purchases minus sales during the period and ignores opening inventory.

3. Freight Cost

Every vendor's freight falls between **0.46% and 0.54%** of purchase cost (median 0.51%). Freight behaves like a flat charge on purchase value: it totals $1.64M, does not separate good vendors from bad ones, and is not a cost-reduction lever.

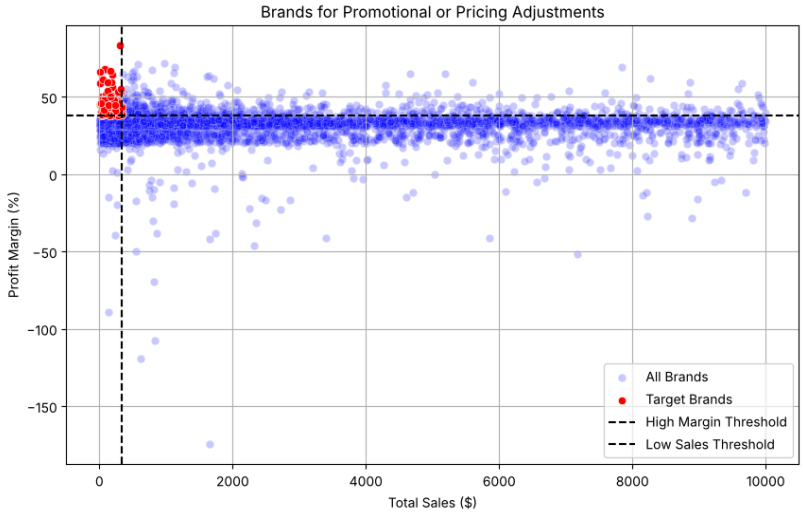
4. Bulk Purchasing

Each brand has exactly one purchase price in the data, so it shows **no evidence of volume discounts**. Brands bought in high volume are cheaper (median $8.60 against $12.71 per unit) because they are lower-priced products, not because of bulk pricing. Testing for discounts would need order-level prices from the raw purchases table.



5. Low-Sales, High-Margin Brands

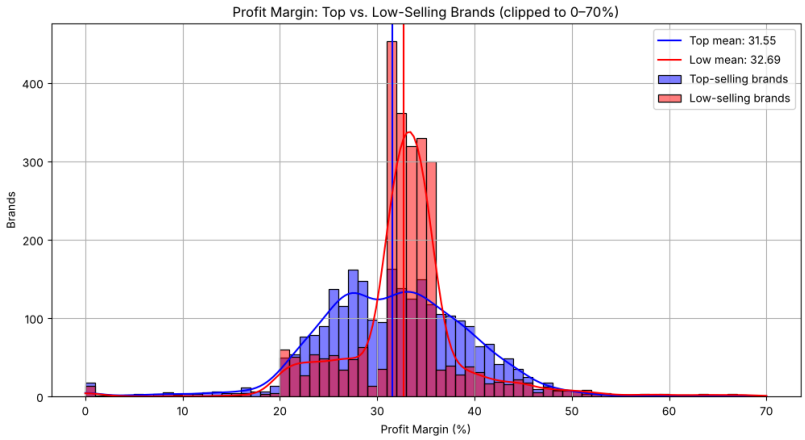
134 brands combine sales of $342 or less with margins of 38% or more. Together they sold $21,206 and earned $9,560 in gross profit, under 0.01% of the total. One margin point on the top 10 brands (about $57M in sales) is worth roughly $570K, so pricing effort belongs there.



6. Margins: Top vs. Low-Selling Brands

Group (quartile by sales)	Brands	Mean margin	95% CI
Top-selling	2,629	31.55%	31.23% – 31.87%
Low-selling	2,630	32.69%	32.38% – 33.00%

Welch's t-test rejects equal means ($t = -4.97$, $p < 0.001$), but the gap is only 1.1 percentage points (Cohen's $d = 0.14$). With samples this large, the test detects a difference too small to justify different pricing strategies. The original 41.6% vs. 31.2% gap came from the gross-profit error above.



Recommendations

- **Cut reorder quantities** on the 2,613 slow-moving brands, starting with the three vendors that hold $2.9M.
- **Review or delist** the 178 brands that never sold and the 104 brands sold below cost.
- **Reduce concentration risk** by keeping secondary suppliers for top brands from Diageo, Martignetti, Jim Beam and Pernod Ricard.
- **Focus pricing work on high-volume brands**, where a one-point margin change is worth about $570K, rather than on low-volume niche brands.